

Paid With(out) Purpose: Perceptions, Preferences, and the Meaning of Work

Online Appendix

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(Link to main paper)

Section and equation numbers prefixed “of the paper” refer to the main manuscript. All tables and figures below are numbered A.1, A.2, and so on. Works cited here are listed in the references of the paper.

A Additional Figures and Tables

Table A.1: Correlation matrix of (ex-ante) task meaning ratings

	Alt text	Subj desc	Decoy 1	Decoy 2	Decoy 3	Decoy 4
Alt text	1.0000					
Subj desc	0.5447	1.0000				
Decoy 1	0.4094	0.3085	1.0000			
Decoy 2	0.3489	0.4848	0.4523	1.0000		
Decoy 3	0.4683	0.2163	0.5916	0.3600	1.0000	
Decoy 4	0.3332	0.7141	0.1305	0.4053	0.0552	1.0000

Notes: Correlation matrix of experimental tasks and decoy tasks rated by participants for meaning on 0-100 scale prior to completing work tasks. *Alt text* is alternative text image description task to aid those with visual impairments. *Subj desc* is subjective image description task to aid general research. *Decoy 1* is teaching a class (vaguely defined) on Zoom. *Decoy 2* is completing freelance work on GitHub or a similar platform. *Decoy 3* is providing telehealth services to patients. *Decoy 4* is completing a brand survey on *Prolific* or a similar platform. $n = 387$. All subjects were required to provide all six task meaningfulness ratings to continue in the experiment. Order in which subjects rated tasks for meaningfulness was randomized across subjects.

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Table A.2: Results of meaning on output for AT and SD task pooled sample

	Wordcount		Quality		Time	
Task-round pay w_{it}	0.008 (0.007)	0.006 (0.006)	0.020 (0.017)	0.016 (0.016)	0.012 (0.014)	0.013 (0.014)
Meaningfulness m_{it}	-0.017 (0.015)	-0.004 (0.015)	-0.031 (0.032)	0.004 (0.033)	0.005 (0.030)	0.003 (0.031)
Task difficulty d_{it}	0.003 (0.013)	0.001 (0.012)	-0.035 (0.034)	-0.041 (0.033)	0.097*** (0.026)	0.091*** (0.026)
Observations	1,548	1,548	1,548	1,548	1,548	1,548
DV Mean	13.27	13.27	66.50	66.50	39.91	39.91
Model	OLS	OLS	OLS	OLS	OLS	OLS
Demographics	.	✓	.	✓	.	✓

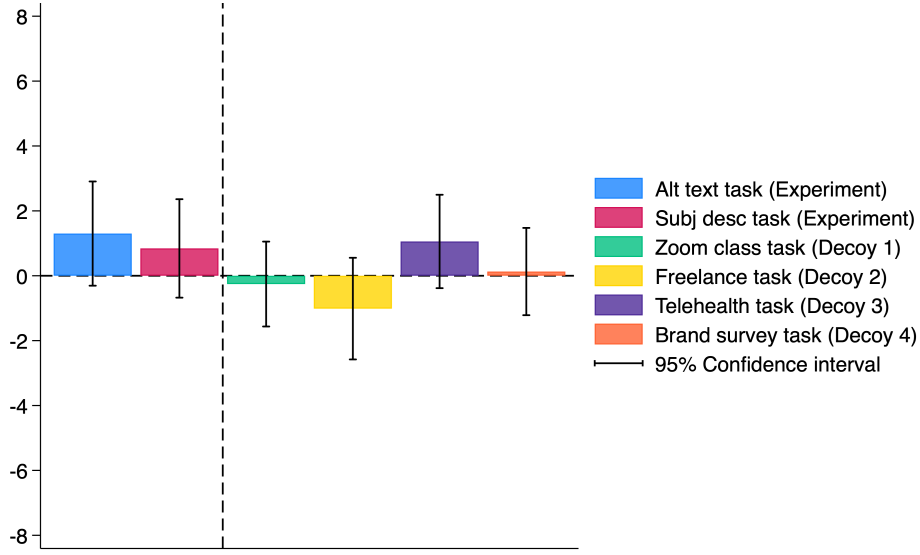
Notes: OLS estimation of subject effort from monetary and nonmonetary work payoffs and effort cost. All regressions include demographic questionnaire order, treatment, and round fixed effects. Task-round pay is amount (in cents) a round offered. Meaningfulness is the subject-reported task meaning measure 0-100 for the assigned task. Task difficulty is the difficulty rating a subject gave for the assigned task. Outcome variable Wordcount is the number of complete or partial words in a subject's image description. Quality is 0-100 rating of subject's image description calculated using GPT-4o. Time is the number of seconds a subject took to submit their image description in a given round (up to 62 seconds when the round auto-advanced). Demographic controls include age, gender, race, employment status, college education, and baseline mood. Standard errors are clustered at the subject level. *** p<0.01, ** p<0.05, * p<0.1.

Table A.3: Effects of meaning on participant satisfaction by choice behavior

Outcome variable: subject endline satisfaction for rounds 5-8						
	Full sample		Non-pay max; non-lucky		Pay maximizers	
$\sum_{r \in [5,8]} w_{itr}$	0.007 (0.016)	0.010 (0.016)	0.001 (0.022)	-0.000 (0.022)	-0.008 (0.027)	-0.017 (0.027)
$\sum_{r \in [5,8]} m_{itr}$	0.086*** (0.013)	0.077*** (0.014)	0.067*** (0.015)	0.061*** (0.016)	0.100*** (0.021)	0.085*** (0.020)
$\sum_{r \in [5,8]} d_{itr}$	-0.058*** (0.011)	-0.055*** (0.011)	-0.075*** (0.015)	-0.073*** (0.015)	-0.041** (0.018)	-0.038** (0.017)
Observations	387	387	226	226	119	119
DV Mean	73.88	73.88	73.77	73.77	72.21	72.21
Demographics	.	✓	.	✓	.	✓

Notes: OLS estimation of participant satisfaction from monetary and nonmonetary work payoffs and effort cost. Outcome variable is 0-100 measure of subject satisfaction with their performance in the four assigned task rounds 5-8. All regressions include demographic questionnaire order, treatment assignment, and number of AT assigned rounds fixed effects. Only meaning, pay, and difficulty from assigned task rounds 5-8 included in sums. Non-pay maximizer; non lucky subsample is subjects whose work choice behavior in rounds 1-4 exhibited losses in both potential pay and potential meaning. Pay maximizers subsample is subjects whose work choice behavior in rounds 1-4 exhibited only losses in potential meaning but never in potential pay. Lucky individuals are those whose choices never exhibited losses in either potential pay or potential meaning. Demographic controls for age, gender, race, employment status, education, and baseline mood are included in the second column of each subsample analysis. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Figure A.1: Task meaningfulness rating differences (After working - before working)



Notes: Within-subject change in meaningfulness ratings (ex-post minus ex-ante, on 0–100 scale) for the AT task, SD task, and four decoy tasks. Error bars show 95% confidence intervals. Differences are near zero and statistically insignificant for both experimental tasks, indicating no evidence of “learning-by-doing” effects on perceptions of meaning.

Alt text: Bar chart of the within-subject change in meaningfulness ratings (ex-post minus ex-ante), with 95 percent confidence interval error bars, for the two experimental tasks and four decoy tasks; all bars are near zero.

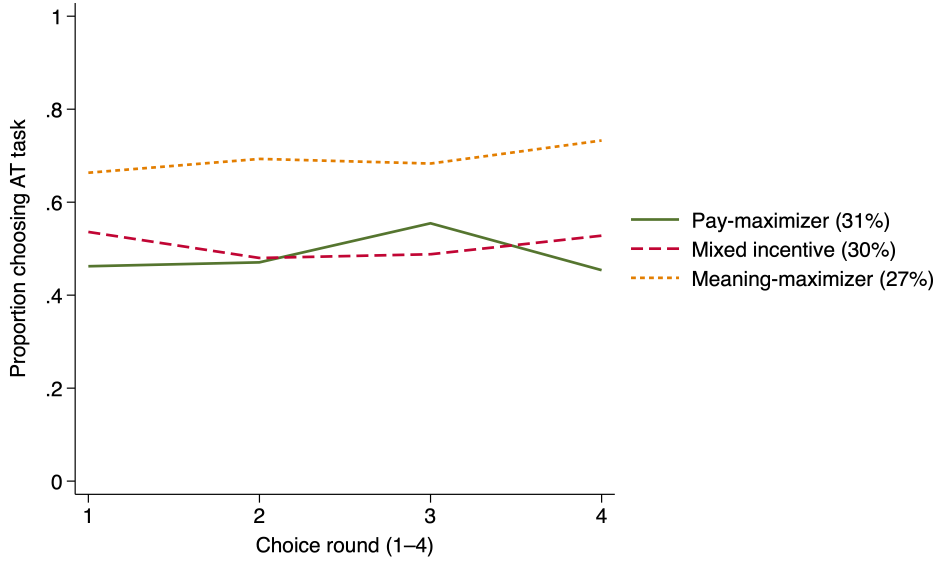
B Conceptual Framework: Derivations

This appendix gives the full statement of the framework summarized in Section 2 of the paper.

I make several key assumptions in this model. First, I assume that all workers have full information¹ on the benefits and costs of their work when making their choices (this means that the model explicitly shuts down “learning-by-doing” whereby workers gain new information about their work in the course of its completion). Second, I assume that all work is completed independently and non-collaboratively, such that no social utility from co-working can be gained by workers on any task. Third, I differentiate the work being completed from volunteering and other altruistic tasks; work explicitly includes a paid wage and a non-zero opportunity cost for lost leisure time. Finally, I assume that all workers are *capable* of providing the minimum required effort on all tasks, such that their failure to do so cannot be interpreted as a lack of skill but instead as a conscious choice.

¹Here I make a key distinction between worker *awareness* and worker *information*. If a worker gains information about their work, they learn something previously unknown about the task they may or will be asked to complete. If a worker gains awareness, however, they are not given any new external information about the task and instead only place more weight on information they already knew about the task.

Figure A.2: Proportion choosing the AT task by choice round and incentive type



Notes: Each line shows the average fraction of workers who chose the AT task in choice rounds 1–4, separately for pay-maximizers (31%), mixed-incentive workers (30%), and meaning-maximizers (27%). Lucky workers (who never faced a meaningful tradeoff) are excluded. Choices are stable across rounds, consistent with fixed underlying preferences rather than learning or fatigue.

Alt text: Line chart of the fraction of workers choosing the alternative text task across choice rounds 1 to 4, plotted as separate lines for pay-maximizers, mixed-incentive workers, and meaning-maximizers; each line is roughly flat across rounds.

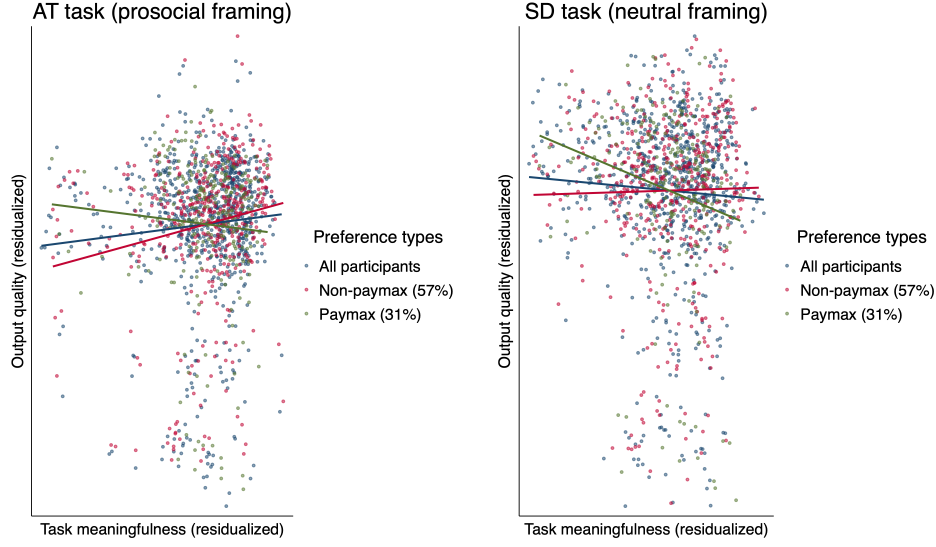
B.1 Intensive Margin of Labor Supply

Following Cassar and Meier (2018), I model the worker’s problem with an additive utility function over the monetary payoffs of work Y , the nonmonetary benefits of work M , and the cost of effort C , each non-decreasing in effort, with the intrinsic motivation of nonmonetary benefits modeled separately from the preference for leisure time.

In this context, I fix a worker’s *perception* of work meaningfulness m as the only nonmonetary benefit of work² that will enter into M . Importantly, as in Cassar and Meier (2018), I model the meaning generated from work as a *flow* rather than a *stock* based on both the base value of m and a worker’s choice of effort e . This means that work meaning is not static and must be generated through the exertion of effort, with greater fixed work meaningfulness

²Previous work has modeled multiple dimensions of work meaning, but here I will collapse all potential sources of work meaning into a single term. This term m can be thought of as the dot product of the fixed characteristics of work and the weights a worker assigns to each of those features.

Figure A.3: Residualized scatter: task meaningfulness and output quality, by task type and incentive type



Notes: Both panels plot residualized output quality (GPT-4o rating) against residualized task meaningfulness after partialling out fixed pay, treatment, difficulty, demographics, and round fixed effects. Left panel: AT (prosocial framing) task. Right panel: SD (neutral framing) task. Lines show OLS fits for all participants (navy), non-pay-maximizers (cranberry), and pay-maximizers (forest green).

Alt text: Two-panel residualized scatterplot of output quality against task meaningfulness (left panel: AT task; right panel: SD task), each with fitted lines for all participants, non-pay-maximizers, and pay-maximizers.

and/or work effort leading to greater generated meaning³. This worker's utility function is:

$$U = Y(w, e) + M(m, e) - C(e) \quad (1)$$

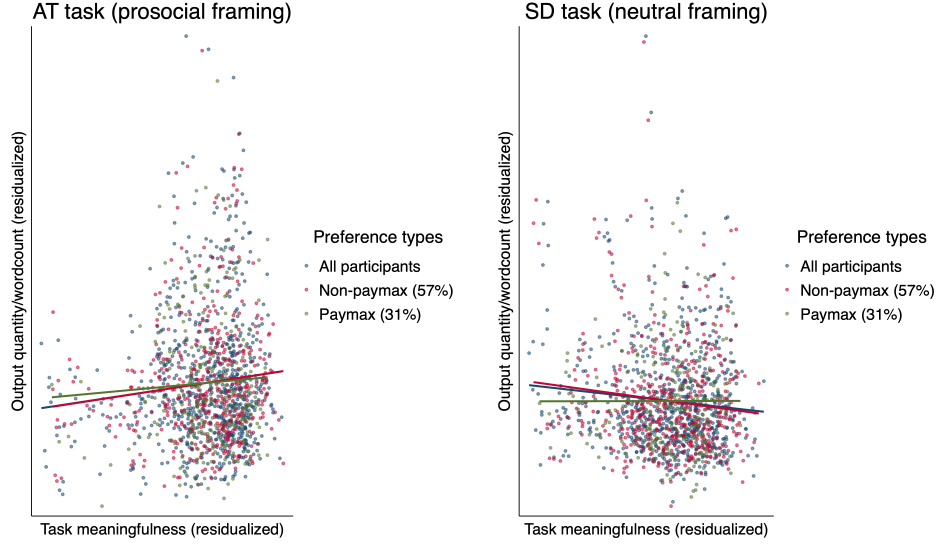
where w is the wage offered for completing the task and m is the meaningfulness of the task. If a worker is optimal, her choice of effort e^* will be that which maximizes her net benefit from providing effort:

$$e^* \equiv \arg \max_e Y(w, e) + M(m, e) - C(e) \quad (2)$$

In this setting, I will assume that monetary and nonmonetary benefits of work enter positively into a worker's utility function and that workers face a cost of expending leisure time.

³I assume exerting no effort on work or finding work entirely non-meaningful should generate no nonmonetary benefit, $M(0, e) = 0$, $M(m, 0) = 0$, while exerting no effort or facing a wage of \$0 should generate no monetary benefit, $Y(0, e) = 0$, $Y(w, 0) = 0$. Conditional on exceeding these thresholds, I assume $\frac{\partial M}{\partial m} |_{e>0} > 0$; analogously, a higher wage generates a larger monetary benefit: $\frac{\partial Y}{\partial w} |_{e>0} > 0$.

Figure A.4: Residualized scatter: task meaningfulness and output quantity, by task type and incentive type



Notes: Both panels plot residualized wordcount against residualized task meaningfulness after partialling out fixed pay, treatment, difficulty, demographics, and round fixed effects. Left panel: AT (prosocial framing) task. Right panel: SD (neutral framing) task. Lines show OLS fits for all participants (navy), non-pay-maximizers (cranberry), and pay-maximizers (forest green).

Alt text: Two-panel residualized scatterplot of output quantity (wordcount) against task meaningfulness (left panel: AT task; right panel: SD task), each with fitted lines for all participants, non-pay-maximizers, and pay-maximizers.

B.2 Extensive Margin of Labor Supply

Now I consider a worker who faces a two-dimensional choice of the task $t \in T$ she will complete as well as the effort e_t she will exert on that task. If this worker is also rational and seeks to maximize her utility, she will choose the task with the highest net benefit at the optimal level of effort for that task. The nature of the experiment motivates considering only a sophisticated decision-maker, one able to consider the hypothetical levels of effort e^t she would exert on all tasks available to her rather than applying an interim, task-invariant level of effort to the choice. I can thus write her two-step decision of effort level and work task, respectively, as follows:

$$e_t^* \equiv \arg \max_{e_t} Y(w_t, e_t) + M(m_t, e_t) - C(e_t) \quad \forall t \in T \quad (3)$$

$$t^* \equiv \arg \max_{t \in T} U(e_t^*, m_t, w_t) \quad (4)$$

Again, I assume the worker has full information about the benefits and costs of providing effort on all tasks $t \in T$, such that she does not learn anything new about her available choices by selecting (or not selecting) a certain task. In the subsequent experiment, workers will be

Table A.4: Ratings of task meaningfulness by worker incentive type

	Alternative text		Subjective description	
Meaning-maximizer	8.70*** (2.33)	10.46*** (2.50)	7.46*** (2.79)	10.17*** (2.95)
Mixed incentive-type	-3.18 (2.56)	-1.33 (2.70)	-1.82 (2.63)	0.89 (2.81)
Leave-out mean	79.53	77.65	72.82	69.70
Demographics	✓	✓	✓	✓
<i>Lucky</i> dropped	.	✓	.	✓
Observations	387	345	387	345

Notes: OLS estimation of 0-100 reported ex-ante work meaningfulness for AT and SD tasks on dummies for observed worker incentive types. Outcome variable is 0-100 measure of meaningfulness reported ex-ante by participants. Pay-maximizers are those whose choices of work in rounds 1-4 never sacrificed potential pay but sacrificed potential meaning. Mixed incentive-types are those whose choices of work sacrificed both potential pay and meaning. Meaning-maximizers are those whose choices of work never sacrificed meaning and always potential pay. Lucky individuals are those whose choices never sacrificed either incentive. Columns (1) and (3) use both pay-maximizer and lucky individuals as leave-out group; columns (2) and (4) drop lucky individuals and use only pay-maximizers as leave-out group. All regressions include demographic question order and treatment assignment fixed effects and demographic controls for age, gender, race, employment status, education, and baseline mood. Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

presented with training rounds for all experimental tasks prior to task or effort selection, giving them an opportunity to consider the optimal level of effort they may exert on all tasks. Workers will also make a selection between only two experimental tasks, reducing cognitive costs of considering many hypothetical tasks simultaneously. Finally, main outcomes from the experiment will be generated only from rounds where workers are (randomly) *assigned* a particular task; rounds where workers make choices of tasks are used only to calibrate worker preference types.

B.3 Worker Incentive Types

The framework so far treats workers as valuing monetary and nonmonetary benefits equally. I now allow workers to differ in their preference for work meaning, holding the relationship between effort and generated meaning (M) fixed across workers, so that at all levels of effort one worker may value the meaning he generates more or less than another. I model the *preferences* of these workers, the relative weight they place on either type of incentive, with the scaling parameter $\phi \in [0, 1]$ as follows:

$$U = \phi Y(w_t, e_t) + (1 - \phi)M(m_t, e_t) - C(e_t) \quad (5)$$

The resulting characterization of workers into *pay-maximizers* ($\phi = 1$), *mixed incentive-types* ($\phi \in (0, 1)$), and *meaning-maximizers* ($\phi = 0$) is shown in Table A.7 and allows me to make testable hypotheses about worker behavior across types.

Hypothesis 1: A worker's incentive type can be identified through (repeated) observation of their choice of task $t^* \in T$ if w_t, m_t are precisely known for all $t \in T$. A worker who demonstrates $t \succ t'$ if $w_t > w_{t'}$ but $m_{t'} > m_t$ is identified to have $\phi = 1$ and thus is a

Table A.5: Multinomial logit regression of worker incentive type on observables

	Type = <i>paymax</i>			Type = <i>meaningmax</i>		
	MLogit	MLogit	MLogit	MLogit	MLogit	MLogit
AT Task Rating	0.007 (0.007)	0.004 (0.007)	0.004 (0.007)	0.033*** (0.010)	0.034*** (0.011)	0.034*** (0.011)
SD Task Rating	-0.006 (0.006)	-0.004 (0.007)	-0.004 (0.007)	0.007 (0.008)	0.009 (0.008)	0.010 (0.009)
Age		0.020* (0.011)	0.020* (0.011)		0.025** (0.012)	0.025** (0.011)
Female		-0.015 (0.273)	-0.003 (0.274)		-0.479* (0.290)	-0.466 (0.291)
White		-0.630 (0.461)	-0.625 (0.466)		-0.497 (0.522)	-0.488 (0.524)
Black		-0.970* (0.533)	-0.985* (0.538)		-0.265 (0.573)	-0.285 (0.579)
Full-time employed		0.243 (0.322)	0.231 (0.320)		0.336 (0.321)	0.315 (0.320)
Part-time employed		0.070 (0.374)	0.058 (0.374)		-0.124 (0.390)	-0.141 (0.388)
No college		0.250 (0.353)	0.257 (0.354)		0.542 (0.355)	0.548 (0.355)
Baseline mood		0.167 (0.178)	0.174 (0.179)		-0.067 (0.192)	-0.052 (0.192)
Treatment			-0.141 (0.266)			-0.274 (0.283)
<i>Lucky</i> dropped	✓	✓	✓	✓	✓	✓
<i>N</i>	345	345	345	345	345	345

Notes: Multinomial logit estimation of worker incentive type on demographic observables, awareness treatment assignment, and demographic questionnaire order (not shown; not significant). The two sets of columns report the pay-maximizer and meaning-maximizer coefficients from the same three specifications, so the sample is common to all six columns. Leave-out group mixed incentive-types. Pay-maximizers are those whose choices of work in rounds 1-4 never sacrificed potential pay but sacrificed potential meaning. Mixed incentive-types are those whose choices of work sacrificed both potential pay and meaning. Meaning-maximizers are those whose choices of work never sacrificed meaning and always potential pay. Lucky individuals who cannot be categorized are dropped from the sample. Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

pay-maximizer. A worker who demonstrates $t \succ t'$ if $m_t > m_{t'}$ but $w_{t'} > w_t$ is identified to have $\phi = 0$ and thus is a *meaning-maximizer*. A worker who demonstrates other behavior is identified to have $\phi \in (0, 1)$ and thus is a *mixed incentive-type*. Furthermore, workers are heterogeneous in their incentive types.

Hypothesis 2: Meaning-maximizers and mixed incentive-types will increase their level of effort on tasks that provide higher meaning (m) while pay-maximizer types will not. With $\phi = 1$, the marginal benefit of generating additional work meaning by increasing effort, $(1 - \phi) \frac{\partial M}{\partial e}$, is zero (across all tasks). Thus, pay-maximizing workers should not increase their effort level in response to a higher base level of work meaningfulness m_t . Similarly, meaning-maximizing individuals with $\phi = 0$ should not increase their effort level in response to a higher wage w_t , as $\phi \frac{\partial Y}{\partial e} = 0$. However, meaning-maximizers and mixed incentive-types face a non-zero marginal benefit of generating additional work meaning for higher base levels of work meaningfulness based on their values of $\phi \in [0, 1)$ and thus should increase their

Table A.6: Awareness treatment effects on ex-ante task meaningfulness ratings

	Alternative text		Subjective description	
	Non-pay-max	Pay-max	Non-pay-max	Pay-max
Awareness treatment	3.73 (2.64)	-2.87 (3.28)	7.26** (2.98)	-2.10 (3.77)
Control mean	78.30	78.30	71.08	71.37
Demographics	✓	✓	✓	✓
<i>Lucky (not characterized)</i> dropped	✓	✓	✓	✓
Observations	226	119	226	119

Notes: OLS ITT estimates of awareness treatment on ex-ante task meaningfulness ratings (0–100), estimated separately by worker incentive type. Lucky (not characterized) workers are excluded from all columns as their incentive type cannot be identified. Non-pay-max columns restrict to mixed incentive-type and meaning-maximizing workers ($n = 226$); pay-max columns restrict to pay-maximizing workers ($n = 119$). Dependent variable is subject’s ex-ante rating of each experimental task framing. Demographic controls include age, gender, race, employment status, college education, and baseline mood. This regression has one observation per subject; standard errors robust to heteroskedasticity (HC1). *** $p < 0.01$. ** $p < 0.05$. * $p < 0.1$.

Table A.7: Worker Incentive Types

Observed Work Choice Behavior	Worker Incentive Type	Values of ϕ
$t \succ t'$ if $w_t > w_{t'}$ but $m_{t'} > m_t$	<i>Pay-maximizer</i>	$\phi = 1$
$t \succ t'$ if $m_t > m_{t'}$ but $w_{t'} > w_t$	<i>Meaning-maximizer</i>	$\phi = 0$
<i>Other behavior</i>	<i>Mixed incentive</i>	$\phi \in (0, 1)$

effort level in response to an increase in m_t across tasks.

B.4 Worker Awareness

Finally, I consider that workers may be more or less aware of the value they place on work meaning. Two workers may share a meaning production function $M(\cdot)$ and meaning weight ϕ but differ on the degree to which they are *aware* of their own preferences (or *see meaning* to borrow the language of Ashraf et al. 2025). To model this, I denote the meaningfulness of work m_t as a function of an awareness parameter $\delta \in [0, \infty)$ such that $m_t = m_t(\delta)$ with $m'_t(\delta) > 0$, so that increased worker awareness leads to increased ratings of task meaningfulness. Importantly, a worker’s perception of meaning m_t increases with δ while preference type (ϕ) is independent of awareness.

Thus, the true utility function faced by workers of various incentive types is

$$U = \phi Y(w_t, e_t) + (1 - \phi) M(m_t(\delta), e_t) - C(e_t) \quad (6)$$

Hypothesis 3: A worker’s meaningfulness rating m_t is dependent on the worker’s awareness δ of how strongly they value the meaning of work. A worker with more awareness, all

else equal, should thus have a higher meaningfulness rating of a task than a worker with low awareness: $m_t(\delta) > m_t(\delta')$ given $\delta > \delta'$. However, a worker’s preferences for generated monetary and nonmonetary benefits of work, ϕ , should not depend on worker awareness and thus should be robust to changes in δ .

C Additional Design and Sample Details

Subject Pool

This experiment was carried out on Prolific in June and October 2025 with responses recorded through Qualtrics. All participants reside in the United States, speak fluent English, are at least 18 years of age, give active consent to participate in the experiment, and have not participated in any pilot versions of the experiment. Workers are told that the experiment is investigating choices over various work tasks and will take them roughly 20 minutes to complete. Subjects are told that data from the experiment will be used to study how choices of work are made and that they will be asked to provide descriptions of a series of pre-selected images (a task domain akin to Burbano 2016). Participants typically complete the experiment in between 15 and 20 minutes, with a base participation fee of \$4 offered to coincide with the platform’s expectation of \$12 hourly compensation. All features of the experiment were approved by the UC Davis Institutional Review Board (IRB) prior to any data collection (IRB Protocol # 2225085-2).

Demographic information is obtained directly from Prolific (age, sex, race and ethnicity, employment status, student status, nationality, country of residence, languages spoken, and number of previous studies), matched to survey results through a de-identified platform ID, and supplemented with in-survey questions on education, political preferences, baseline mood, gender, and previous altruistic behavior. To avoid any priming of behavior as a result of these questions, participants are randomly assigned to complete this demographic questionnaire either at the baseline or endline of the experiment and a question order fixed effect is included in all regression specifications (though does not generally appear statistically significant). The order of demographic questions presented to participants is also randomized within the survey block.

Controls for being Asian, US-born, a student, or previous charitable giving or volunteering are ultimately not included in the regression specifications in the paper due to a lack of variation in the experimental sample: 95.6% of subjects are US-born, 95.6% are non-Asian, 87.6% are not students, and 83.2% report having engaged in altruistic behavior (volunteering or charitable donation) during the past year. Demographic characteristics for workers are summarized in the full-sample column of Table A.8. Demographic controls included in all subsequent regressions are age, gender, race (indicators for White and Black), employment status, education level, and baseline mood.

Task Framings, Training, and Experimental Flow

Appendix Figure A.1 plots the difference between ex-ante and ex-post meaningfulness ratings of both AT and SD experimental tasks, as well as four decoy tasks, demonstrating that

Figure A.5: Experimental task framings

Subjective Description	Alternative Text
We are conducting research on how subjects describe various images in both human and non-human settings.	We also hope to provide written descriptions alongside images in future experiments to help identify them for participants with visual impairments.
You will have one minute (60 sec) to write a 5-word (minimum) description of the image presented to you.	You will have one minute (60 sec) to write a 5-word (minimum) description of the image presented to you.
Descriptions should be both detailed and accurate, capturing key elements of the image such as setting and main subject.	Descriptions should encompass any notable details or features of the image.
By selecting this option, you consent to your description being used for this research.	By selecting this option, you consent to your description being used for this purpose.

Notes: All participants were shown this graphic (in non-randomized left-to-right order) prior to rating or completing tasks.

Alt text: Side-by-side graphic comparing the two image-description task framings shown to participants, the subjective description (SD) task and the alternative text (AT) task, each presented with its instructions and its stated purpose or mission.

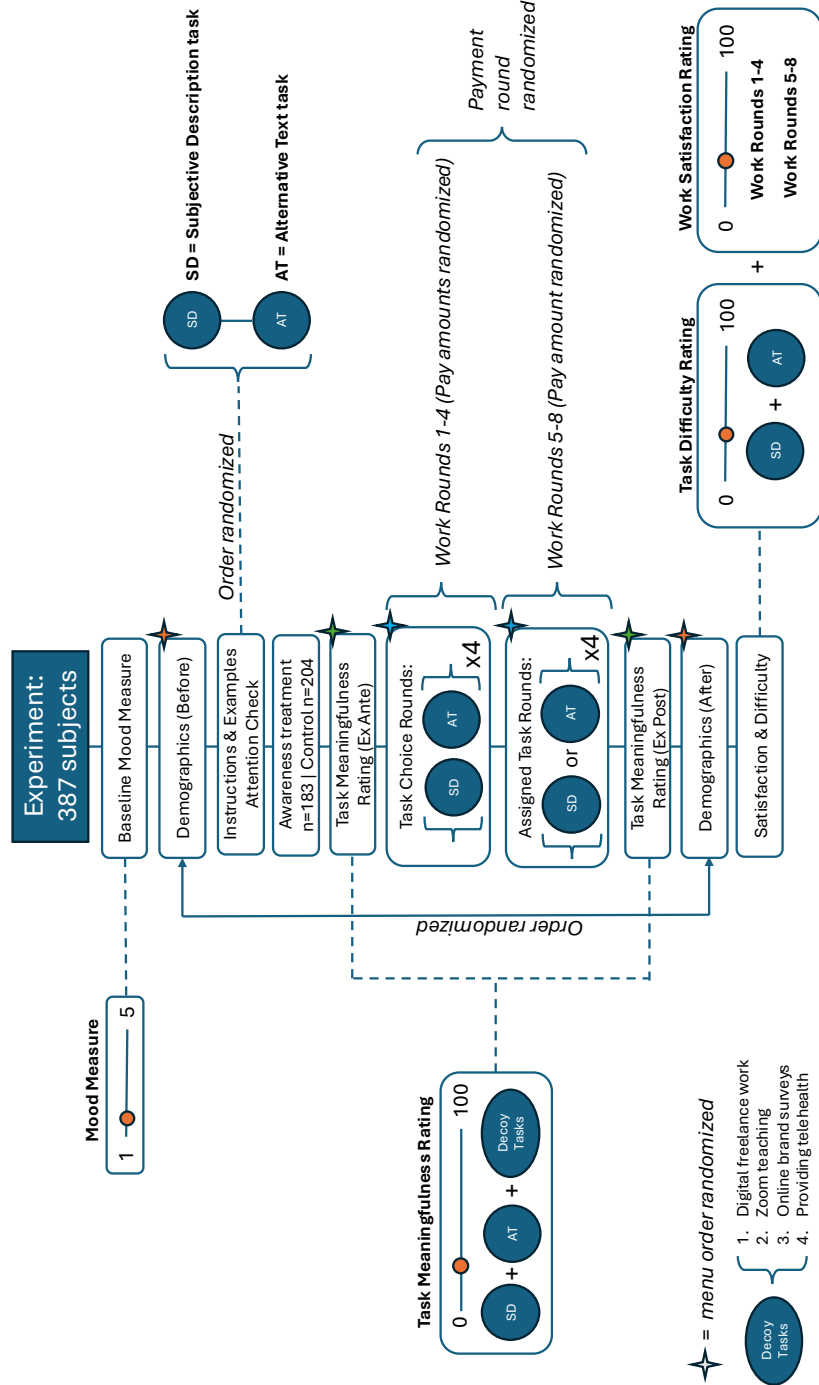
perceptions of meaning did not change as a result of the experiment and thus no “learning-by-doing” took place. Meaningfulness ratings of both experimental tasks are correlated at 0.54 before completing work (Table A.1) and 0.47 after completing work.

Several factors ensure that participants are comfortable and familiar with the experimental tasks they will be asked to complete. Subjects are first given examples of five “good” alternative text and subjective descriptions of two example images (in randomized order). These descriptions were generated using GPT-4o by supplying it with the example images and the exact task instructions (see Appendix E). Following the description of work tasks, participants face an attention check screen that requires them to correctly identify the two tasks they will complete in the experiment, Alternative Text and Subjective Description. The order of options a subject sees for the attention check is also randomized.

A timeline of the experiment is shown in Figure A.6. Stars indicate a block in which the order of objects shown to subjects is randomized. Demographic questions are presented, in random order, to all subjects either before or after completing the work component of the experiment. The screen asking subjects to rate the two experimental tasks and four decoy tasks for meaningfulness presents these options in randomized and re-randomized order at the baseline and endline of the experiment, respectively. The task choice rounds present subjects with a choice of the AT or SD task, in randomized order left-to-right on the screen. Finally, the assigned task rounds present subjects randomly with either the AT or SD task to complete in a round. All images for which subjects are asked to write descriptions, along with screenshots of the rating screen, the training rounds, the attention check, the mood measure, and the choice and assigned work rounds, are reproduced in the working paper version.

For all work rounds, workers see a clock counting down their remaining time until submission, a reminder of the task they are being asked to complete in that round, the image

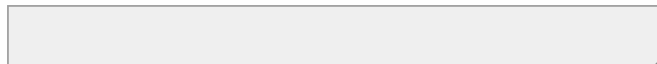
Figure A.6: Timeline of Experiment



Alt text: Flow diagram of the experiment timeline, read left to right, showing the sequence of stages from consent and randomization through task meaningfulness ratings, training, choice rounds, assigned work rounds, and baseline/endline questionnaires; stars mark stages presented to subjects in randomized order.

Figure A.7: Light-touch awareness treatment

Please list three online tasks you would find *meaningful* to complete, along with your reasoning for each. This will help improve future research designs.



Alt text: Screenshot of the light-touch awareness treatment screen, which prompts participants to brainstorm and write about types of online work they would personally find meaningful.

they are asked to describe, and a blank text box. Workers are told they have 60 seconds to submit their descriptions, but the round automatically submits workers’ responses after 62 seconds. This slight window accounts for delays in the survey platform. Importantly, a worker who does not manually submit their answer and instead has their work round automatically submit still has their answer recorded up until the moment of submission, such that exhausting all allotted time does not result in a worker having a missing submission.

Awareness Treatment and Balance

At the beginning of the experiment, workers are randomized into either a treatment or control group. The treatment group is shown the light-touch awareness intervention shown in Figure A.7 prior to the ex-ante task meaningfulness ratings while the control group is shown no additional screen. Participants in the treatment group have unlimited time to complete the open-ended brainstorming question but are required to write at least one character before submitting and continuing on to the remainder of the experiment. These subjects show strong engagement with the brainstorming task despite its lack of financial incentive or bearing on the remainder of the experiment. Subjects write roughly 33 words on average (with a median of 30) and give answers that demonstrate they take the question seriously⁴.

Table A.8 shows that demographic variables are well-balanced across the treatment and control groups, with only participant baseline mood ratings showing a mean difference across the two groups. All regressions thus include demographic controls for baseline mood, along with age, gender, and level of education.

Construction of Outcome Variables

The three output measures are defined in Section 3.3 of the paper. This section gives the detail omitted there: how the quality ratings were produced, examples of how the model scored descriptions, and the measurement error the ratings carry.

For one of the experimental images, the model rated “A turtle laying on the sand with water from the ocean behind him” 100 and “Sea turtle resting on a sandy beach with waves in the background.” 95, while “at the shore sleeping turtle” received 45, “it is a beach” 20, “Turtle”

⁴Subjects systematically use words like ‘help’, ‘helps’, and ‘helping’ when describing tasks they would find meaningful.

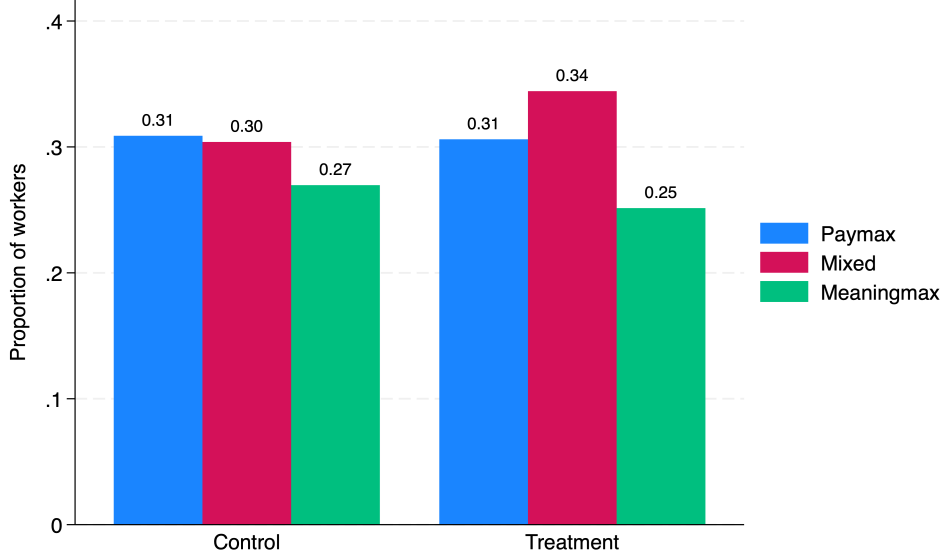
Table A.8: Summary statistics and balance of demographic variables by treatment and control group

	Full sample	Treatment	Control	Mean difference
<i>Included in regressions</i>				
Age	40.039 (13.485)	39.295 (13.427)	40.706 (13.534)	-1.411 (1.373)
Female	0.553 (0.498)	0.596 (0.492)	0.515 (0.501)	0.081 (0.051)
White	0.685 (0.465)	0.716 (0.452)	0.657 (0.476)	0.059 (0.047)
Black	0.214 (0.411)	0.180 (0.386)	0.245 (0.431)	-0.065 (0.042)
Full-time employed	0.364 (0.482)	0.328 (0.471)	0.397 (0.490)	-0.069 (0.049)
Part-time employed	0.202 (0.402)	0.197 (0.399)	0.206 (0.405)	-0.009 (0.041)
No college degree	0.240 (0.428)	0.251 (0.435)	0.230 (0.422)	0.021 (0.044)
Baseline mood	4.003 (0.800)	4.087 (0.780)	3.926 (0.812)	0.161** (0.081)
<i>Excluded from regressions</i>				
Asian	0.044 (0.205)	0.044 (0.205)	0.044 (0.206)	-0.000 (0.021)
US-born	0.956 (0.205)	0.945 (0.228)	0.966 (0.182)	-0.020 (0.021)
Student	0.124 (0.330)	0.126 (0.332)	0.123 (0.329)	0.003 (0.034)
Altruistic behavior	0.832 (0.374)	0.836 (0.371)	0.828 (0.378)	0.008 (0.038)
Observations	387	183	204	387

Notes: Summary statistics and mean differences for subjects separated by random assignment to treatment or control group. Subjects were recruited in Summer and Fall 2025. Subjects were required to be 18 years of age, understand English, and live in the United States. Subjects who had participated in earlier pilot versions of the experiment were disqualified from participating. All subjects earned a base pay amount of \$4 in line with Prolific guidelines. The full sample column describes the experimental sample as a whole: the average participant is 40 years of age (ranging from 18 to 83), female, white, full-time employed, not a student, has either volunteered or donated to charity (“Altruistic behavior”) in the past year, and reports being in quite a good mood (4/5) entering the experiment. Standard deviations in parentheses in the first three columns; standard errors of the mean difference in parentheses in the fourth. 183 subjects were assigned to the treatment group and 204 subjects were assigned to the control group. Differences in subject assignment rates are due to disconnection issues on the survey platform and not significant differential attrition by participants. Mean differences presented between treatment and control group. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

10, and “a” 0. Overall the model performs with internal consistency and accuracy, correctly flagging poor quality or nonsense ratings as having low quality and correctly assigning high

Figure A.8: Effect of awareness treatment on worker incentive types



Notes: Graph of proportion of worker incentive types by random assignment to treatment or control. 183 individuals assigned to light-touch awareness treatment condition; 204 assigned to control condition with no additional screen. Differences in treatment assignment due to features of the experimental platform and do not indicate differential attrition by participants. Paymax individuals are those whose choices of work in rounds 1-4 demonstrated no valuation of work meaning; Mixed incentive type individuals are those whose choices were motivated by both pay and meaning; Meaningmax individuals are those whose choices demonstrated no valuation of work pay. P-values of mean differences by treatment and control groups are $p=0.952$, $p=0.398$, $p=0.684$ for pay-maximizers, mixed incentive-types, and meaning-maximizers, respectively.

Alt text: Grouped bar chart comparing the proportion of pay-maximizer, mixed-incentive, and meaning-maximizer workers in the treatment versus control groups; the paired bars are nearly identical across conditions for every type.

quality ratings to apt and detailed descriptions. The model does not simply assign higher ratings to longer descriptions, however, as the correlation between description wordcount and description quality is only 0.43. Thus, the model is correctly identifying quality in the *content* of descriptions rather than simply rating on the *length* of descriptions.

However, as no LLM is (yet) perfect, I assume that the observed output quality rating from GPT-4o is a noisy measure of true output quality, such that $q^{observed} = q^{true} + \eta$. Ideally this noise term η is mean zero with some variance σ_η^2 such that the model does not systematically under- or over-rate descriptions. In practice, the distribution of rating residuals (regressed on task difficulty and demographics) is left-skewed rather than symmetric, driven primarily by a small share of observations ($\approx 4\%$ of the sample) rated exactly zero by the model. These may reflect genuinely poor descriptions or residual API irregularities despite automated retry logic. Importantly, any systematic differences in mean rating levels across task types are absorbed by the task fixed effects included in all regressions, so task-level differential standards do not threaten identification. The remaining concern is whether the zero-rated observations are correlated with the regressors of interest; given their small share, and the close agreement between the OLS quality estimates and fractional logit estimates that model the bounded outcome and its zero boundary directly (Appendix D), they are unlikely to drive the main estimates. Thus, while measurement error in quality ratings may

be non-normal, it should not systematically bias estimates of the impact of work meaning and would at most inflate standard errors on regression coefficients, reducing precision.

On average, subjects provide image descriptions well above the 5-word minimum threshold required for payment. On average subjects write 13.27 words for an image in rounds 5-8 of the experiment, an amount significantly above the 5-word minimum threshold ($p < 0.001$). Subjects spend an average of 40 seconds writing their descriptions, well below the maximum time of 62 seconds available, and have an average quality rating of 67 out of 100. These results suggest that subjects took the work offered to them seriously and provided real effort on these tasks.

Demographic Questionnaire

Subjects were asked four demographics questions, in randomized order, either before or after completing task ratings and work tasks in the experiment: whether they had given a charitable donation of any size to a nonprofit organization in the past year (*yes, no, I don't remember*); whether they had volunteered in their community at least once in the past year (*yes, no, I don't remember*); their maximum education level (*some high school, completed high school, some college, completed college, some graduate level, completed graduate level*); and their gender identity (*male, female, non-binary or other gender*). Information on age, race, employment, and country of origin was obtained directly from *Prolific*. To utilize the most up-to-date and representative gender information for participants, sex information reported by *Prolific* was replaced with gender information elicited from participants during the experiment. Due to a lack of variation (and incentive compatibility), responses for previous volunteering and charitable giving were dropped from the list of covariates for estimation.

Deviations from Preregistration

This study was preregistered in February 2025 on the American Economic Association's Randomized Controlled Trial Registry (AEARCTR-0015326). Since its initial preregistration, some procedural aspects of the project have been updated while its core research question and design have remained the same. On outcomes, the preregistered probability of a subject selecting a particular task in a given round has been adapted into computing the quantity of potential pay and meaning a worker's choices of task sacrifice over four rounds, which is used to classify workers into incentive types; description wordcount and AI-generated rating of description quality remain exactly as preregistered, while the third effort measure has been adjusted from a word-per-second measure of productivity to a standalone measure of round submission time. Secondary preregistered outcomes on worker wellbeing are examined more fully in a companion paper, "What Moves Meaning? Investigating Drivers of an Idiosyncratic Incentive" (SSRN working paper); reported work satisfaction is studied in both papers, entering here only as a proxy for realized utility (Table A.3), and baseline mood is collected as preregistered and used as a demographic control in all regressions. On sample size, the current experiment uses only 387 subjects where the preregistered experiment targeted 2000, mainly because it collapses several planned treatment arms into treatments given to all workers. Rather than half-randomizing workers into rating all experimental tasks for

meaningfulness only before or only after working, all workers now rate tasks both before and after completing them, which allows me to investigate possible “learning-by-doing” effects within-subject (I find no evidence of this). Rather than randomizing workers into receiving or not receiving information on the “mission” of the Alternative Text task and a training round, I now provide *all* workers with both, due to concerns that the task may otherwise be novel and confusing enough that ex-ante ratings of its meaningfulness would be uninformative. Finally, the preregistered version planned for workers to complete five rounds of image description tasks where the current version sees them complete eight. The pilot version of this experiment was approved by the UC Davis IRB under Protocol ID 2225085-1; the updated version was subsequently registered with and approved by the UC Davis IRB under Protocol ID 2225085-2, with all changes to the experimental design properly justified.

D Poisson PPML, Fractional Logit, and Tobit Robustness Estimates

This appendix presents robustness checks on the main-text OLS estimates using estimators chosen to match the distributional features of each outcome variable (see Section 3.3 of the paper). Wordcount is estimated using Poisson PPML (average partial effects reported). Quality, which is bounded on $[0, 100]$ and rescaled to $[0, 1]$ for estimation, is estimated using fractional logit (average partial effects in percentage-point units reported). Submission time, which is censored at the 62-second round limit, is estimated using Tobit with an upper censoring point at 62 seconds (latent-index coefficients reported). The two tables below mirror the worker-level specifications of the paper, Panels B and C of Table 1 and all of Table 2, with the full-sample AT and SD estimates reported in the first panel of each. Panel A of Table 1 is not reproduced here: it estimates the effect of being assigned the task rated more meaningful on average, and serves to show that this design misestimates the effect of meaningful work rather than to measure it. Results are broadly consistent with the OLS estimates throughout. Standard errors are clustered at the participant level.

Table A.9: Results of meaning on output for AT task by incentive type (Poisson/Fracreg/Tobit)

	Wordcount		Quality		Time	
Full sample (n=779)						
Task-round pay $w_{i,AT}$	0.005 (0.008)	0.005 (0.008)	0.000 (0.021)	0.000 (0.021)	0.047* (0.024)	0.047* (0.024)
Meaningfulness $m_{i,AT}$	0.028* (0.015)	0.033** (0.015)	0.045 (0.048)	0.079 (0.048)	0.093** (0.042)	0.074* (0.044)
Task difficulty $d_{i,AT}$	-0.014 (0.014)	-0.014 (0.013)	-0.085** (0.042)	-0.088** (0.041)	0.075** (0.038)	0.067* (0.038)
Mixed incentive types & meaning-maximizers (n=437)						
Task-round pay $w_{i,AT}$	-0.001 (0.011)	-0.005 (0.010)	-0.010 (0.029)	-0.017 (0.029)	0.043 (0.032)	0.042 (0.032)
Meaningfulness $m_{i,AT}$	0.031 (0.019)	0.033* (0.018)	0.124* (0.067)	0.164** (0.067)	0.029 (0.057)	0.019 (0.057)
Task difficulty $d_{i,AT}$	-0.017 (0.015)	-0.016 (0.015)	-0.073 (0.061)	-0.068 (0.054)	0.032 (0.048)	0.025 (0.048)
Pay-maximizers (n=256)						
Task-round pay $w_{i,AT}$	0.004 (0.016)	0.012 (0.015)	0.010 (0.035)	0.013 (0.034)	0.045 (0.045)	0.058 (0.042)
Meaningfulness $m_{i,AT}$	0.015 (0.032)	0.021 (0.032)	-0.101 (0.083)	-0.085 (0.084)	0.171* (0.091)	0.147 (0.092)
Task difficulty $d_{i,AT}$	-0.018 (0.027)	-0.005 (0.024)	-0.091 (0.076)	-0.062 (0.075)	0.082 (0.073)	0.108 (0.071)
Model	Poisson	Poisson	Fracreg	Fracreg	Tobit	Tobit
Demographics	.	✓	.	✓	.	✓

Notes: Robustness estimates. Sample restricted to rounds where the alternative text task was randomly assigned. Wordcount estimated via Poisson PPML (APEs reported). Quality estimated via fractional logit (APEs in percentage-point units reported). Timing estimated via Tobit with upper censoring at 62 seconds (latent-index coefficients reported). All regressions include demographic questionnaire order, treatment, and round fixed effects. Task-round pay is amount (in cents) a round offered. Meaningfulness is the subject-reported task meaning measure 0-100 for the alternative text task. Task difficulty is the difficulty rating a subject gave for the alternative text task. Results separated by observed work choice behavior. Lucky workers (11% of the sample) are excluded from the by-type subgroup regressions but included in the full-sample row. Demographic controls include age, gender, race, employment status, college education, and baseline mood. Standard errors are clustered at the subject level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.10: Results of meaning on output for SD task by incentive type (Poisson/Fracreg/Tobit)

	Wordcount		Quality		Time	
Full sample (n=769)						
Task-round pay $w_{i,SD}$	0.010 (0.010)	0.007 (0.009)	0.035 (0.026)	0.027 (0.024)	-0.014 (0.024)	-0.011 (0.024)
Meaningfulness $m_{i,SD}$	-0.048** (0.020)	-0.030 (0.019)	-0.085** (0.036)	-0.047 (0.037)	-0.068 (0.047)	-0.063 (0.048)
Task difficulty $d_{i,SD}$	0.022 (0.016)	0.018 (0.015)	0.017 (0.037)	0.008 (0.036)	0.158*** (0.038)	0.157*** (0.038)
Mixed incentive types & meaning-maximizers (n=467)						
Task-round pay $w_{i,SD}$	0.014 (0.013)	0.007 (0.012)	0.067** (0.034)	0.044 (0.030)	0.015 (0.028)	0.017 (0.028)
Meaningfulness $m_{i,SD}$	-0.059* (0.031)	-0.037 (0.030)	-0.034 (0.052)	0.012 (0.048)	-0.066 (0.056)	-0.059 (0.056)
Task difficulty $d_{i,SD}$	0.015 (0.019)	0.012 (0.018)	0.067 (0.050)	0.052 (0.046)	0.146*** (0.050)	0.152*** (0.050)
Pay-maximizers (n=220)						
Task-round pay $w_{i,SD}$	0.002 (0.017)	0.007 (0.018)	-0.047 (0.055)	-0.021 (0.050)	-0.028 (0.050)	-0.015 (0.051)
Meaningfulness $m_{i,SD}$	-0.022 (0.025)	-0.000 (0.024)	-0.254*** (0.088)	-0.216** (0.096)	-0.022 (0.084)	0.001 (0.097)
Task difficulty $d_{i,SD}$	0.042 (0.035)	0.037 (0.031)	-0.042 (0.071)	-0.040 (0.068)	0.188*** (0.067)	0.195*** (0.070)
Model	Poisson	Poisson	Fracreg	Fracreg	Tobit	Tobit
Demographics	.	✓	.	✓	.	✓

Notes: Robustness estimates. Sample restricted to rounds where the subjective description task was randomly assigned. Wordcount estimated via Poisson PPML (APEs reported). Quality estimated via fractional logit (APEs in percentage-point units reported). Timing estimated via Tobit with upper censoring at 62 seconds (latent-index coefficients reported). All regressions include demographic questionnaire order, treatment, and round fixed effects. Task-round pay is amount (in cents) a round offered. Meaningfulness is the subject-reported task meaning measure 0-100 for the subjective description task. Task difficulty is the difficulty rating a subject gave for the subjective description task. Results separated by observed work choice behavior. Lucky workers (11% of the sample) are excluded from the by-type subgroup regressions but included in the full-sample row. Demographic controls include age, gender, race, employment status, college education, and baseline mood. Standard errors are clustered at the subject level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

E Use of Large Language Models

This appendix documents the use of large language models (LLMs) in the paper, as required by the journal’s disclosure policy. A single model, OpenAI’s GPT-4o, was used in two places: to generate the output-quality ratings analyzed in the paper, through the chat-completions API, and to produce the example descriptions shown to participants during training, through the ChatGPT web interface. Only the first of these produced data analyzed in the paper.

Output-quality ratings (data analyzed in the paper). The output-quality measure used throughout the paper’s productivity regressions is generated by GPT-4o. For each participant description, the model received the referenced image, the task instructions shown to the participant, and the participant’s text, and returned an integer quality score on a 0–100 scale. Ratings were obtained through separate, independent API calls (temperature = 0.2) so that no rating could be informed by any other, with automatic retry logic for any call that failed to return a valid integer. The verbatim prompt is also reproduced in a footnote in Section 3 of the paper. The system message was: “You are an assistant that rates the quality of a user’s text description of an image on a continuous numerical scale from 0 (terrible) to 100 (excellent). Descriptions should be rated higher if they are written in complete, coherent sentences that clearly and accurately convey visual details. Descriptions consisting mostly of lists of adjectives, sentence fragments, or vague phrases should receive lower scores. Your score should reflect how effectively the description helps someone visualize the image. Please respond with only a single integer between 0 and 100, with no explanation. This rating will be used for research purposes, so be precise and consistent.” The user message was “Rate this text from 0 to 100: [description]”, supplied together with the corresponding image, where [description] is the participant’s submission. The rating script was executed once per data-collection wave, on June 10, 2025 (OpenAI 2025a) and October 20, 2025 (OpenAI 2025b). During development the script was pilot-tested on a subsample of descriptions to identify and fix bugs; those pilot ratings were discarded, and only the two finalized runs cited above produced the ratings analyzed in the paper.

Training examples (experimental materials). During task training, participants were shown example “good” alternative-text and subjective descriptions of two sample images (Section 3 of the paper). These example descriptions were generated with GPT-4o through the ChatGPT web interface (<https://chat.openai.com>) prior to data collection, by supplying the model with the two sample images and the exact task instructions given to participants and asking it to produce a description for each task. They are experimental stimuli shown to participants and are not themselves analyzed as data, so they are disclosed here for transparency but are not cited as a data source.

No generative AI was used to write or substantively rewrite the manuscript.